Introduction

A cell, in biology, is a membrane-bound unit that contains the fundamental molecules of life. The cells of all living organisms create an elegant molecular order within them and pass a pattern of that order on to descent organisms. This creation and duplication of order, out of often disordered surroundings, is unique to a living cell, and the complexity is governed by continual expenditure of energy. A cell is responsible for the everlasting process of abstracting/obtaining energy from their surroundings- either from sunlight, as plants do, or from foodstuffs, as animals do. This is achieved by the ordered structural organization and functions of subcomponents of each cell which will be dealt in detail.

1.1 The unit of biological organisation: The Cell

In 1838, two scientists Theodor Schwann and Matthias Schleiden, were chatting after-dinner about their observations on cells. Schleiden was describing the plant cells with nuclei which made Schwann to recall the similarity of these plant cells to the animal cells which he had studied. The two scientists went immediately to Schwann's lab and examined the slides. Later Schwann published a book on animal and plant cells (Schwann 1839) but did not acknowledge anybody (including Schleiden 1838). He concluded his cell paper as :

i) The cell is the unit of structure, physiology, and organization in living things.

ii) The cell retains a dual existence as a distinct entity and a building block in the construction of organisms.

iii) Cells form by free-cell formation, similar to the formation of crystals (spontaneous generation).

Be it a complex living organism or a non-living matter, they are comprised of several basal units. If you are a living organism, you are based on the single block, the cell. If you are non-living matter like a house, you are based on the single block, the brick. Thus, simply, a cell is a basal building unit of an organism. One of the first major discoveries in biology was made by Robert Hooke (1665). He showed that the partition of the cork (plant tissue) into tiny compartments, and termed them as 'cellulae', or cells. In 1838 Schleiden, a German Botanist and his co-worker Theodor Schwann, have shown that organisms are made up of cells surrounded by a thin layer. However, they failed to correctly explain the generation of cells. Later, in 1857, Rudolph Virchow, a pathologist demonstrated that cells can be formed. only by the division of pre-existing cells. Yet, the question of first cell remains unanswered. Single cells may be adapted to survive in many different types of environments, extending between extremes of cold or heat, living in aerobic or anaerobic conditions, or even surrounded by methane gas. Some single cells can live within other organisms.

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